9.22 (a) For TE mode, or the magnetic multipole field, the electric field in the cavity is given by Eq. (9.116), with $g_l(kr) = j_l(kr)$. This solution is transverse and thus has only θ and ϕ components. The metallic boundary with infinite conductivity requires that the θ and ϕ components of the electric field vanishes at $r = a$, which means $j_l(ka) = 0$. Let x_{ln} be the nth solution to $j_l(x) = 0$, then the characteristic frequencies are given by $\omega_{lmn} = \frac{c}{a} x_{ln}$, which are independent of m.

Similarly, for TM mode, the magnetic and the electric fields are given by Eq. (9.118), again with $f_l(kr) = j_l(kr)$. From Eq. (10.60),

$$\nabla \times \left(f_l(r) \vec{X}_{lm} \right) = \frac{i \vec{n} \sqrt{l(l+1)}}{r} f_l(r) Y_{lm} + \frac{1}{r} \frac{\partial}{\partial r} \left[r f_l(r) \right] \vec{n} \times \vec{X}_{lm},$$

where $\vec{X}_{lm} = \vec{L} Y_{lm} / \sqrt{l(l+1)}$. We can see that the transverse components of the electric field are given by the second term. Again, we require the transverse components vanish at the boundary, which leads to $\left.\frac{\partial}{\partial r} \left[r j_l(kr) \right] \right|_{r=a} = 0$. Let x'_{ln} be the nth solution to the equation $\frac{d}{dx} [x j_l(x)] = 0$, then the characteristic frequencies are $\omega_{lmn} = \frac{c}{a} x'_{ln}$, independent of m.

(c) The electric field in the TE mode is given by

$$\vec{E}^{(M)}_{lm,n} = Z_0 j_l(kr) \vec{L} Y_{lm},$$

and magnetic field is

$$\vec{H}^{(M)}_{lm,n} = -\frac{i}{kZ_0} \nabla \times \vec{E}^{(M)}_{lm,n} = -\frac{i}{k} \sqrt{l(l+1)}\, \nabla \times \left(j_l(kr) \vec{X}_{lm} \right)$$

$$= -\frac{i}{k} \sqrt{l(l+1)} \left[\frac{i \vec{n} \sqrt{l(l+1)}}{r} j_l(kr) Y_{lm} + \frac{1}{r} \frac{\partial}{\partial r} \left[r j_l(kr) \right] \vec{n} \times \vec{X}_{lm} \right]$$

$$= \vec{n} \frac{l(l+1)}{kr} j_l(kr) Y_{lm} - \frac{i}{kr} \frac{\partial}{\partial r} \left[r j_l(kr) \right] \vec{n} \times \vec{L} Y_{lm}, \quad \text{with } k = x_{ln}/a$$

Similarly for TM mode.

$$\vec{H}^{(E)}_{lmn} = j_l(k'r)\,\vec{L}\,Y_{lm}(\theta,\phi)$$

$$\vec{E}^{(E)}_{lmn} = \frac{iZ_0}{k'}\nabla\times\vec{H}^{(E)}_{lmn} = \frac{iZ_0}{k'}\sqrt{l(l+1)}\;\nabla\times\left[\,j_l(k'r)\,\vec{X}_{lm}\right]$$

$$= \frac{iZ_0}{k'}\sqrt{l(l+1)}\left[\,\frac{i\vec{n}\sqrt{l(l+1)}}{r}\,j_l(k'r)\,Y_{lm} + \frac{1}{r}\frac{\partial}{\partial r}\left[r\,j_l(k'r)\right]\vec{n}\times\vec{X}_{lm}\,\right]$$

$$= -\vec{n}\,\frac{Z_0\,l(l+1)}{k'r}\,j_l(k'r)\,Y_{lm} + \frac{iZ_0}{k'r}\frac{\partial}{\partial r}\left[\,r\,j_l(k'r)\right]\vec{n}\times\vec{L}\,Y_{lm}\,.$$